

Velocity sorting biases the equilibrium in Array-RQMC for binary-collision models

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Abstract

Array-RQMC aligns the sorted states of many trajectories with a low-discrepancy point set. We consider it for binary-collision models of the Kac type, ordering one system's particles by their own state and pairing by that order. We give a condition whose violation certifies that this adaptation moves the equilibrium, and show that velocity-based orderings violate it. Averaging over the scattering angle gives a fourth-moment identity in which the pairing rule enters through one scalar cross term, extremised by equal speeds or extremes. Any departure from its state-independent value moves the fourth moment on the first collision, at every even $N \geq 4$ and any symmetric law common to all pairs not concentrated on the quarter turns. Within this adaptation the alignment producing the variance reduction is what biases the equilibrium. We prove this for the sorted and extremal rules and measure it for block and curve rules, not for every velocity-dependent key. The obstruction is to the adaptation, not to Array-RQMC itself: sorting independent copies and leaving their partners alone is untouched. In experiments variance reduction is delivered while mean squared error worsens, replicates agreeing closely on an incorrect answer. A one-pass diagnostic condemns a pairing before any trajectory runs; a corollary exhibits an admissible class. For the sorted rule the mechanism is a monotone energy cascade whose drain is total: under the uniform law the two fastest almost surely carry all the energy in the limit, with uniform phase, which fixes the stationary law; which particles they are keeps changing.

Key words. randomized quasi-Monte Carlo, Array-RQMC, Kac model, invariant measure, particle methods, collisional kinetics

AMS subject classifications. 65C05, 65C35, 82C40, 11K45

1 Introduction

Array-RQMC works by sorting. At each step the states of many trajectories are sorted, the sorted states are aligned with a low-discrepancy point set, and every trajectory is advanced using its assigned point; how much variance the method saves depends on the sorting key. For binary-collision models the obvious key is a particle's own velocity. We show that it moves the equilibrium the simulation is meant to sample, and that the alignment saving the variance is the same structure that moves it.

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Array-RQMC is due to L’Ecuyer, Lécot and Tuffin [12], and was anticipated for Markov chains in one dimension by Lécot and Tuffin [11]. It has produced substantial variance reductions across a range of problems, and the empirical study of sorting rules by L’Ecuyer et al. [13] showed that the choice of sorting key matters a good deal, with the Hilbert-curve ordering of Gerber and Chopin [5] performing well in several dimensions. He and Owen [6] treat space-filling curves for sampling; Ho and Owen [7, 8] apply Array-RQMC to grid-free solvers, where a walk on spheres with a Hilbert sort reduces variance by factors running from tens to thousands. General background on randomized quasi-Monte Carlo is in Niederreiter [16], Dick and Pillichshammer [4] and Owen [17].

Part of why the method works is understood. Randomized quasi-Monte Carlo works best when the integrand is dominated by the low-order terms of its ANOVA decomposition, made precise by the effective dimension of Caflisch, Morokoff and Owen [1] and the mean dimension of Liu and Owen [14]. Sorting the states lets a chain present the point set with a low-dimensional integrand at each step.

Collisional kinetics looks like a natural target for the method. The trajectories are long, the per-step dimension is small, and the quantity of interest is an average over many particles: the regime in which Array-RQMC has done well elsewhere. The obvious sorting key involves the particle velocities, since it is the velocities that the collision operator acts on.

The idea is not new. Coulibaly and Lécot [3] gave a quasi-Monte Carlo particle simulation of the Kac model in which the velocities are reordered at every step and the coordinates of a (t, m, s) -net then index into the sorted array to choose which velocities combine; a companion scheme for a linear Boltzmann equation is in [10]. They proved convergence as the particle number grows, with a bound their own analysis calls pessimistic, and their experiments showed the scheme outperforming Monte Carlo even so. They also noted, in closing, that quasi-random points are delicate to use because the particles must be reordered, that many orderings are available in more than one dimension, and that some of them fail to improve on Monte Carlo. That the ordering can decide whether the method helps was therefore known. What was missing is a reason, and a way to tell in advance which orderings are safe. The route is also the one a production code reaches for: the implementation of Section 7, written before the obstruction was understood, orders its particles by a curve through phase space and pairs adjacent ranks.

We give a condition whose violation certifies that a pairing rule moves the equilibrium, a one-pass test for it, and the class of orderings that survive. The velocity-based orderings do not survive: this adaptation moves the equilibrium, and the reason is structural rather than a matter of implementation. Section 4 establishes an identity for a Kac binary-collision model: averaged over the scattering angle, the fourth moment after one collision depends on the pairing rule only through the single scalar $C = \sum_{(a,b)} v_a^2 v_b^2$. That scalar takes a definite value under any pairing chosen independently of the state, and by a rearrangement argument it is largest when partners have equal speeds and smallest when they are extreme. Any rule whose cross term departs from the exchangeable value therefore moves the fourth moment on the first collision, so the equilibrium is not preserved, for every even particle number $N \geq 4$. The identity survives replacing the uniformly resampled scattering angle by any symmetric law shared by all pairs, provided that law is not concentrated on the quarter turns that leave Q fixed, with the drift scaled by a computable factor, which shrinks the drift for forward-peaked laws and enlarges it for others, so small-angle scattering does not evade the conclusion. A law that varies from pair to pair, as a physical cross-section does, falls outside the reduction; Remark 4.6 records why.

The consequence is specific to one way of using the sort. Where the order of the particles is what selects their collision partners, aligning them with a low-discrepancy point set requires ordering by the state, and an ordering that carries velocity information can then bias the cross

term. Corollary 4.10 establishes that it does for the sorted and extremal rules; for the block and curve rules the bias is measured rather than proved. Within that adaptation the alignment producing the variance reduction and the bias destroying the equilibrium are the same structure, and among the rules examined here no choice of sorting key separates them. Where instead the sort assigns low-discrepancy points to copies of a chain and leaves each copy's transition law alone, none of this applies; Definition 2.2 draws the line. Section 6 bears this out: the variance reduction is delivered as advertised while the mean squared error is monotonically worsened, with no interior optimum to tune towards.

Two features make the failure difficult to notice, and we think they are worth as much attention as the theorem. The conserved moments are pinned by construction, so a calculation can be wrong by a factor of several hundred in the fourth moment while every routine conservation check reports perfection. And the biased scheme is the more reproducible of the two: its replicates agree to within half a percent of each other while sitting far from the truth, by a margin that the replicate spread itself would call overwhelming. Tight agreement between replicates, in this setting, is a symptom rather than a reassurance.

The proof is constructive enough to be useful. The cross term is computable in $O(N)$ operations from a single state, before any trajectory is run, and comparing it against its exchangeable value tests a proposed scheme in one pass. Section 8 sets out what the test does and does not establish, and identifies the sorting strategies that remain admissible: grouping by a key that carries no velocity information, as the binary-collision schemes in standard use [18] do, leaves the equilibrium intact.

What is established and what is not should be stated plainly. Sections 2 to 4 are proved. Section 5 is proved apart from one hypothesis, displayed as (10). Proposition 5.5 reduces that hypothesis, at every even $N \geq 6$ at once, to a geometric condition on one pair of functions of one variable, displayed as (11); that condition is verified with margin rather than proved, and nothing else in the reduction is assumed. The section also carries one illustration that is measurement and is marked as such. Section 6 is measured, with the precision of each quantity determined by its sampling behaviour rather than assumed. Section 9 settles where the chain goes. For the sorted rule the energy of the slowest particles can only fall, at every collision and along every path, and for any scattering law with a density spread over more than a quarter turn the drain is total: almost surely the two fastest end with the whole energy, though which particles those are keeps changing, and under the uniform law the stationary law follows by arithmetic. What remains open is the length of the moment hierarchy and how much velocity information a sort may carry.

2 The Kac model and its equilibrium

Kac [9] introduced the following caricature of a spatially homogeneous gas. Its approach to equilibrium has been studied in detail: McKean [15] obtained the rate for the associated kinetic equation, and Carlen, Carvalho and Loss [2] determined the spectral gap of the master equation. Both concern the unbiased dynamics; the question here is what a state-dependent pairing rule does to the invariant measure itself. Let N particles carry scalar velocities $v = (v_1, \dots, v_N)$. A collision selects a pair of indices and rotates the two velocities in their coordinate plane, leaving the other components alone. The energy of the pair is conserved exactly, so the total energy is conserved, and after normalising the state moves on the sphere

$$\mathcal{S} = \left\{ v \in \mathbb{R}^N : \sum_i v_i^2 = N \right\}. \quad (1)$$

Throughout, N is an *even* integer. This is a hypothesis and not a convenience: a collision step matches the N indices into $N/2$ disjoint pairs, and a perfect matching exists only for even N . Every statement below is quantified over even N .

We consider the step in which all $N/2$ pairs of a matching collide simultaneously, which is the form a particle code uses. For a pair (a, b) put $R^2 = v_a^2 + v_b^2$ and set

$$v_a \leftarrow R \cos(\phi + \theta), \quad v_b \leftarrow R \sin(\phi + \theta), \quad (2)$$

where $(v_a, v_b) = R(\cos \phi, \sin \phi)$ before the collision and θ is drawn independently for each pair. The update rotates the pair through θ . When θ is uniform on $[0, 2\pi)$ the new phase is uniform and independent of ϕ , which is Kac's collision; Section 4 keeps the rotation and lets θ carry an arbitrary distribution symmetric about zero. The rotation is the model throughout: replacing ϕ by θ instead would agree with (2) in law for the uniform case and differ for every other, and it is not used here.

Let σ denote the uniform measure on $\mathcal{S}$. It is the equilibrium of the model, and its low-order structure is available in closed form: the normalised squares v_i^2/N are marginally $\text{Beta}(\frac{1}{2}, \frac{N-1}{2})$, whence

$$\mathbb{E}_\sigma[v_i^4] = \frac{3N}{N+2}, \quad \mathbb{E}_\sigma[Q] = \frac{3N^2}{N+2}, \quad \mathbb{E}_\sigma[Q_6] = \frac{15N^3}{(N+2)(N+4)}, \quad (3)$$

where $Q(v) = \sum_i v_i^4$ and $Q_6(v) = \sum_i v_i^6$. These exact values matter in what follows: they let bias be computed rather than estimated.

Definition 2.1. A *pairing rule* assigns to each state a perfect matching of $\{1, \dots, N\}$, possibly at random. The rule is *state-independent* if the matching is chosen without reference to v , whether deterministically or from a distribution not depending on v , and *state-dependent* otherwise.

The object of this paper is a particular way of bringing Array-RQMC to a collision model, and it should be named, because it is not the only one.

Definition 2.2. By a *state-dependent collision pairing rule* we mean the adaptation in which the particles of one system are ordered by their own state and that order is used to choose which particles collide with which. This is to be distinguished from Array-RQMC as ordinarily formulated, in which the objects sorted are independent copies of a Markov chain and the sort decides only which low-discrepancy point each copy receives, leaving the transition law of every copy untouched.

The distinction is the whole of what follows. Sorting copies of a chain is a statement about the assignment of randomness and changes no transition law. Sorting the particles of one system to choose collision partners changes the transition kernel itself, and it is that modified kernel which the next section shows cannot preserve σ . Nothing here rules out an Array-RQMC construction for a Kac system that leaves the collision partners alone.

The scheme of Coulibaly and Lécot [3] orders the particles of one system and lets that order select the velocities that combine, so it is of the first kind; but it draws the two indices of each collision independently, so one velocity may enter several collisions and another none. The rules of Definition 2.1 partition the particles into disjoint pairs, as a binary-collision code does [18], and the results below are stated for them; they say nothing about sampling with replacement. Their convergence theorem is also asymptotic in the particle number at a fixed time, whereas the question here is what happens to the invariant measure at the particle numbers a simulation actually runs, so the two results do not meet.

The distinction is exactly the one that matters, and one direction is immediate.

Proposition 2.3. *Under any state-independent pairing rule, σ is invariant.*

Proof. The update (2) applied across a matching acts on v as a product of plane rotations in disjoint coordinate planes. Within the plane of a pair the rotation carries the incoming phase ϕ to θ , so the angle turned through is $\theta - \phi$, which is not a function of the sampled randomness alone. It is here that uniformity is used: since the uniform law on the circle is invariant under shifts, $\theta - \phi$ is uniform and independent of ϕ , hence independent of v . (For the general scattering law of Lemma 4.5 the pair is rotated through θ directly and no such step is needed.)

Write the step as $v \mapsto R(\omega)v$, where ω collects the matching and the turned angles; by the previous paragraph ω is independent of v whenever the matching is. For $v \sim \sigma$ and any measurable E , conditioning on ω and using the rotation invariance of σ gives $\Pr(R(\omega)v \in E) = \mathbb{E}_\omega[\sigma(R(\omega)^{-1}E)] = \sigma(E)$. $\square$

Proposition 2.3 makes no use of randomness in the matching. A single fixed matching, held for all time, preserves σ as surely as a freshly drawn one, though of course it fails to mix. Neither determinism nor structure in the pairing can therefore be the source of any failure of invariance; only dependence on the state can be.

A state-dependent rule is a piecewise rotation: the sphere partitions into regions on which the matching is constant, and a different rotation is applied on each. The following records when such a map preserves σ . It is not used later, but it locates the mechanism and explains the exceptional case $N = 2$, where only one matching exists and no state dependence is possible.

Proposition 2.4. *Let $\{A_i\}$ be a countable measurable partition of $\mathcal{S}$, let R_i be rotations, and let $T(v) = R_i v$ for $v \in A_i$. Then $T_*\sigma = \sigma$ if and only if $\{R_i(A_i)\}$ is again a partition of $\mathcal{S}$ up to σ -null sets.*

Proof. For measurable E , since each R_i preserves σ ,

$$T_*\sigma(E) = \sum_i \sigma(A_i \cap R_i^{-1}E) = \sum_i \sigma(R_i A_i \cap E) = \int_E m \, d\sigma, \quad m = \sum_i \mathbf{1}_{R_i A_i}.$$

Because $\sum_i \sigma(R_i A_i) = \sum_i \sigma(A_i) = 1$, the multiplicity function satisfies $\int m \, d\sigma = 1$ for every such T . Hence $T_*\sigma = \sigma$ if and only if $m = 1$ almost everywhere. $\square$

3 Pairing rules

The rules below are used throughout. All but the first are state-dependent. Ranks are always taken with respect to speed $|v_i|$, ties broken arbitrarily; on $\mathcal{S}$ ties occur with probability zero.

- **Random.** A uniform random perfect matching, drawn afresh each step and independent of the state.
- **Sorted.** Order the particles by $|v_i|$ and pair adjacent ranks: $(1, 2), (3, 4), \dots$. Partners have nearly equal speeds.
- **Extremal.** Order by $|v_i|$ and pair rank i with rank $N + 1 - i$. Partners have maximally unequal speeds.
- **Block of width w .** Order by $|v_i|$, cut the ordered list into consecutive blocks of w , and draw a uniform random matching inside each block. Taking $w = 2$ recovers the sorted rule and $w = N$ recovers the random rule, so w interpolates between them while changing nothing else.

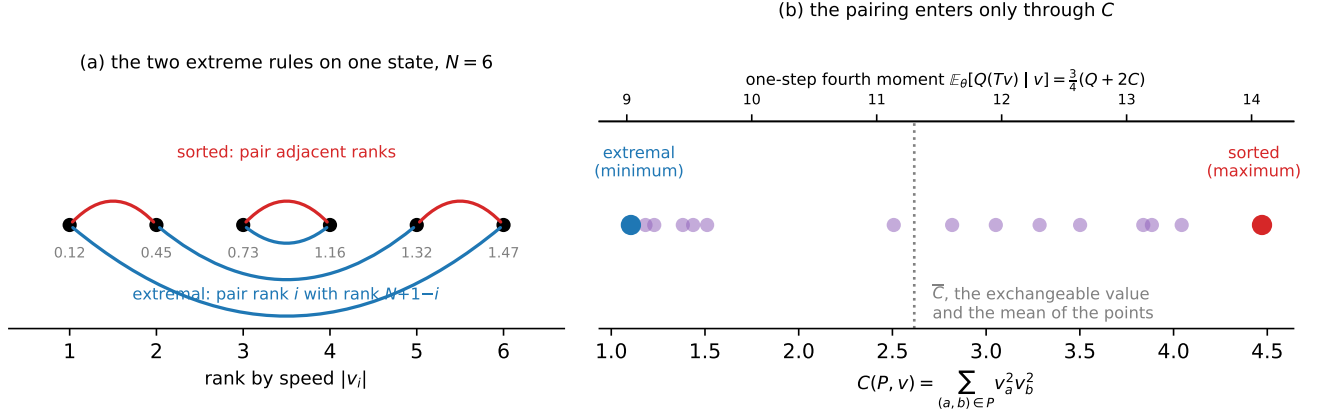


Figure 1: The mechanism, on one state of $N = 6$ particles drawn from σ . (a) The six speeds in rank order, with the sorted rule pairing adjacent ranks drawn above the axis and the extremal rule, pairing rank i with rank $N+1-i$, drawn below it. (b) All fifteen perfect matchings placed on the C axis. By Lemma 4.1 the sorted matching gives the largest value and the extremal the smallest; by Lemma 4.7 the exchangeable value $\bar{C}$ is their mean, marked by the dotted line. The top axis carries the one-step fourth moment that Lemma 4.2 assigns to each, so a rule that systematically selects matchings away from $\bar{C}$ moves $\mathbb{E}_\sigma[Q]$ by the corresponding amount. Generated by `kac_fig1_spectrum.v1_0_0.py`.

- **Cell.** Partition the particles by an auxiliary coordinate carrying no velocity information, such as spatial position, and draw a uniform random matching inside each cell. This is the pairing used by binary-collision schemes of Takizuka–Abe type [18].
- **Curve.** Order the particles by the index of a space-filling curve evaluated at $(x_i, \Phi(v_i/v_{\text{th}}))$, where x is position, Φ is the standard normal distribution function and v_{th} is the thermal speed, then pair adjacent ranks. This is the rule an Array-RQMC implementation uses, following the Hilbert-curve sort of Gerber and Chopin [5] and the comparisons of L’Ecuyer et al. [13].

The sorted, extremal and block rules are idealisations that isolate one effect. The cell and curve rules are the ones a practitioner would actually write down, and Sections 7 and 8 return to them.

4 The obstruction

The argument has three ingredients: a rearrangement lemma about matchings, an identity for the one-step fourth moment, and the observation that the two meet in a single scalar. Figure 1 shows all three on one state.

4.1 A rearrangement lemma for matchings

Lemma 4.1. *Let $x_1 \leq x_2 \leq \dots \leq x_{2n}$ be real numbers and, for a perfect matching P of $\{1, \dots, 2n\}$, put $S(P) = \sum_{(a,b) \in P} x_a x_b$. Then S is maximised by the adjacent matching $\{(1, 2), (3, 4), \dots, (2n-1, 2n)\}$ and minimised by the extremal matching $\{(1, 2n), (2, 2n-1), \dots, (n, n+1)\}$. If the x_i are distinct and $n \geq 2$, each extremiser is unique and every other matching gives a value strictly between the two.*

Proof. Let P maximise S and suppose 1 is matched to j and 2 to k with $j, k \geq 3$ and $j \neq k$. Replacing these two pairs by $(1, 2)$ and (j, k) changes S by

$$(x_1x_2 + x_jx_k) - (x_1x_j + x_2x_k) = (x_2 - x_j)(x_1 - x_k) \geq 0,$$

because $x_2 \leq x_j$ and $x_1 \leq x_k$. Hence some maximiser matches 1 with 2; removing that pair leaves a sorted list of $2n - 2$ entries and the claim follows by induction. The increment is strictly positive unless $x_2 = x_j$ or $x_1 = x_k$, which gives uniqueness when the x_i are distinct.

For the minimum, let P minimise S and suppose 1 is matched to j and $2n$ to k with $j, k \notin \{1, 2n\}$. Replacing these pairs by $(1, 2n)$ and (j, k) changes S by

$$(x_1x_{2n} + x_jx_k) - (x_1x_j + x_{2n}x_k) = (x_{2n} - x_j)(x_1 - x_k) \leq 0,$$

because $x_{2n} \geq x_j$ and $x_1 \leq x_k$. Induct as before. $\square$

For $n = 2$ the lemma reads $x_1x_2 + x_3x_4 \geq x_1x_3 + x_2x_4 \geq x_1x_4 + x_2x_3$, which the two displayed identities verify directly. We note that this is not the classical rearrangement inequality, which concerns two sequences and the sum of their products under a permutation; the statement needed here is about partitioning one sequence into pairs, and the exchange argument above is what supplies it.

4.2 The one-step fourth moment

Lemma 4.2. *Let T denote one collision step (2) applied across a matching P , each angle being uniformly distributed on $[0, 2\pi)$. No independence across pairs is required. The angles are drawn independently of P and of v . Then for any P , state-dependent or not,*

$$\mathbb{E}_\theta[Q(Tv) \mid v] = \frac{3}{4} \left(Q(v) + 2C(P, v) \right), \quad C(P, v) = \sum_{(a,b) \in P} v_a^2 v_b^2. \quad (4)$$

Proof. A pair contributes $R^4(\cos^4 \theta + \sin^4 \theta)$ to $Q(Tv)$, and $\mathbb{E}[\cos^4 \theta] = \mathbb{E}[\sin^4 \theta] = 3/8$, so its conditional expectation is $\frac{3}{4}R^4$. Summing over pairs and expanding, $\sum_P (v_a^2 + v_b^2)^2 = \sum_i v_i^4 + 2 \sum_P v_a^2 v_b^2$. $\square$

Identity (4) is the engine of everything below, and the point to notice is what it does not contain. The pairing rule enters the one-step fourth moment only through the single scalar C ; every other quantity in (4) depends on the state alone. Whatever elaborate structure a sorting scheme imposes, its entire effect on this moment is summarised by one number.

The restriction to uniformly resampled angles is inessential.

Remark 4.3. A rule may draw its matching at random given the state, and Proposition 5.3 uses one that does. Conditioning first on v and then on P and averaging in that order replaces $C(P, v)$ in (4) by $\mathbb{E}_\theta[C(P, v) \mid v]$, the average over whatever distribution the rule assigns to P at that state. Every statement below that takes an expectation over a randomised rule is to be read this way.

Remark 4.4. Only the marginal law of each angle is used. Writing $Q(Tv)$ as a sum over the pairs of P and taking expectations term by term needs linearity alone, so (4) holds whatever the joint law of the family $(\theta_{ab})_{(a,b) \in P}$, provided each θ_{ab} is uniform. That matters in the present setting: the angles supplied by a randomized quasi-Monte Carlo point set are individually uniform and deliberately correlated, and the lemma covers them.

Lemma 4.5. *Suppose instead that the collision rotates the pair through an angle θ drawn from one distribution, symmetric about zero and common to every pair, the draws themselves being independent, so that $(v_a, v_b) = R(\cos \phi, \sin \phi) \mapsto R(\cos(\phi + \theta), \sin(\phi + \theta))$, and set $\kappa = \mathbb{E}[\cos 4\theta]$. Then*

$$\mathbb{E}_\theta[Q(Tv) \mid v] = \kappa Q(v) + (1 - \kappa) \frac{3}{4} (Q(v) + 2C(P, v)). \quad (5)$$

Proof. For a single pair, $v_a^4 + v_b^4 = R^4(1 - \frac{1}{2} \sin^2 2\phi) = R^4(\frac{3}{4} + \frac{1}{4} \cos 4\phi)$, so that $R^4 \cos 4\phi = 4(v_a^4 + v_b^4) - 3R^4$. After the rotation the same expression holds with ϕ replaced by $\phi + \theta$, and averaging over θ symmetric about zero annihilates the term in $\sin 4\theta$ and multiplies the term in $\cos 4\phi$ by κ . Hence

$$\mathbb{E}_\theta[v_a'^4 + v_b'^4] = \frac{3}{4} R^4 + \frac{\kappa}{4} R^4 \cos 4\phi = (1 - \kappa) \frac{3}{4} R^4 + \kappa (v_a^4 + v_b^4).$$

Summing over pairs and using $\sum_P R^4 = Q + 2C$ gives (5). $\square$

Uniform resampling is the case $\kappa = 0$, where (5) reduces to (4), and $\kappa = 1$ is the degenerate case in which the collision cannot change Q . For every $\kappa < 1$ the pairing still enters only through C , and the coefficient multiplying it is simply scaled by $1 - \kappa$. The scaling is an attenuation only when κ is positive: small-angle scattering, for which κ is close to one, shrinks the effect described below without removing it, while a law with $\kappa < 0$ has $1 - \kappa > 1$ and enlarges it. Either way the pairing enters only through C , and the obstruction is unaffected in kind.

4.3 The exchangeable value

Remark 4.6. A common law across pairs is needed, and this is the boundary of the single-scalar reduction. If the angle law varies with the pair, so that $\kappa_{ab} = \mathbb{E}[\cos 4\theta_{ab}]$ depends on (a, b) , the same calculation gives

$$\mathbb{E}_\theta[Q(Tv) \mid v] = \sum_{(a,b) \in P} \left(\kappa_{ab} (v_a^4 + v_b^4) + (1 - \kappa_{ab}) \frac{3}{4} R_{ab}^4 \right), \quad R_{ab}^2 = v_a^2 + v_b^2,$$

which depends on which κ is attached to which pair and so is not a function of Q and C alone; states agreeing in (Q, C) but differing in this quantity exist. The reduction to a single scalar, and with it Theorem 4.9, is therefore a statement about a common scattering law. A physical cross-section, whose angle law depends on the relative speed of the colliding pair, lies outside it.

Lemma 4.7. *For an exchangeable rule, meaning one that draws the matching uniformly at random independently of the state, the conditional expectation of C given v is*

$$\overline{C}(v) = \frac{N^2 - Q(v)}{2(N - 1)}. \quad (6)$$

Substituting (6) into (4) and taking expectations over σ gives an affine map on $m = \mathbb{E}_\sigma[Q]$,

$$\Phi(m) = \frac{3}{4} \cdot \frac{m(N - 2) + N^2}{N - 1}, \quad (7)$$

whose slope $\frac{3}{4}(N - 2)/(N - 1)$ is less than one for every N , and whose unique fixed point is $m^* = 3N^2/(N + 2) = \mathbb{E}_\sigma[Q]$. For any state-independent rule, exchangeable or not, $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$.

Proof. By exchangeability, $\mathbb{E}[C \mid v] = (N/2) \mathbb{E}[v_a^2 v_b^2]$ for a uniformly chosen pair (a, b) , which equals $(2(N-1))^{-1} \sum_{i \neq j} v_i^2 v_j^2$. Using $\sum_i v_i^2 = N$ this is $(N^2 - Q)/(2(N-1))$, giving (6), and substitution yields (7). Solving $\Phi(m) = m$ gives $m(N+2) = 3N^2$. The identification of m^* with $\mathbb{E}_\sigma[Q]$ is (3).

For the last claim, let P_0 be any fixed matching. Since σ is invariant under permutations of the coordinates, $\mathbb{E}_\sigma[v_a^2 v_b^2]$ takes the same value for every pair $a \neq b$, so summing over the $N/2$ pairs of P_0 gives $\mathbb{E}_\sigma[C(P_0, \cdot)] = (N/2) \mathbb{E}_\sigma[v_1^2 v_2^2] = (N^2 - \mathbb{E}_\sigma[Q])/(2(N-1)) = \mathbb{E}_\sigma[\overline{C}]$. A state-independent rule is a mixture of fixed matchings, so the same holds for it. $\square$

Remark 4.8. The distinction matters. A fixed matching is state-independent by Definition 2.1, but for it $\mathbb{E}[C \mid v] = C(P_0, v)$, which by Lemma 4.1 may be the largest or the smallest cross term available at that state rather than the average. What survives for every state-independent rule is the equality of σ -expectations, and that is all Theorem 4.9 uses.

Two remarks. First, Q is a random variable and the fixed point in (7) is of the induced map on its expectation, not of a scalar recursion; the map closes on the mean because it is affine in Q , which is what makes the argument legitimate. Second, Lemma 4.7 is consistent with Proposition 2.3, as it must be: a state-independent rule leaves $\mathbb{E}_\sigma[Q]$ where it found it.

4.4 The theorem

Theorem 4.9. *Let P be a pairing rule for which $\mathbb{E}_\sigma[C(P(v), v)] \neq \mathbb{E}_\sigma[\overline{C}(v)]$. Then σ is not invariant. If the left-hand side is the larger, one step from σ strictly increases $\mathbb{E}_\sigma[Q]$; if it is the smaller, the step strictly decreases it. Under a scattering law with parameter $\kappa < 1$ the same conclusion holds with the change in $\mathbb{E}_\sigma[Q]$ scaled by $1 - \kappa$.*

Proof. Integrating (4) against σ gives $\mathbb{E}_\sigma[Q(Tv)] = \frac{3}{4}(\mathbb{E}_\sigma[Q] + 2\mathbb{E}_\sigma[C])$. By Lemma 4.7, replacing $\mathbb{E}_\sigma[C]$ by $\mathbb{E}_\sigma[\overline{C}]$ returns $\mathbb{E}_\sigma[Q]$ exactly. The right-hand side is strictly increasing in $\mathbb{E}_\sigma[C]$, so $\mathbb{E}_\sigma[C] \neq \mathbb{E}_\sigma[\overline{C}]$ forces $\mathbb{E}_\sigma[Q(Tv)] \neq \mathbb{E}_\sigma[Q]$ with the stated sign. Were σ invariant the two would agree. The final claim follows the same way from (5), whose dependence on C carries the factor $1 - \kappa$. $\square$

The hypothesis is an inequality between integrals, which is weaker than pointwise domination and is all the proof uses. Two ratios follow from it and must be kept apart for the rest of the paper. Write

$$r(v) = \frac{C(P(v), v)}{\overline{C}(v)} \quad \text{and} \quad r_\sigma = \frac{\mathbb{E}_\sigma[C]}{\mathbb{E}_\sigma[\overline{C}]},$$

the first read from a single state and the second an average over the equilibrium law. Theorem 4.9 constrains r_σ . What a practitioner can compute from a state in hand is $r(v)$, and the two are not interchangeable: a fixed matching is state-independent, so $r_\sigma = 1$ exactly by Lemma 4.7, while $r(v)$ at a given state may be the largest or the smallest value available there.

Corollary 4.10. *For even $N \geq 4$, the measure σ is invariant under neither the sorted nor the extremal rule. Sorted pairing strictly increases $\mathbb{E}_\sigma[Q]$ on the first step, and extremal pairing strictly decreases it. For $N = 2$ there is a unique matching, no rule can depend on the state, and σ is preserved.*

Proof. Apply Lemma 4.1 with $x_i = v_i^2$. The sorted rule realises the maximum of C over matchings pointwise and the extremal rule the minimum. Under σ the v_i^2 are almost surely distinct and, for $N \geq 4$, at least three matchings exist, so by the uniqueness clause the maximum strictly exceeds

the average over matchings almost surely, and likewise the minimum falls strictly below it. Taking expectations gives $\mathbb{E}_\sigma[C] \neq \mathbb{E}_\sigma[\overline{C}]$ with the stated sign in each case, and Theorem 4.9 applies. The case $N = 2$ is Proposition 2.3. $\square$

The two directions describe different failures. Pairing partners of equal speed drives the fourth moment up, which under a fixed second moment means mass accumulating near zero while a thin population carries the energy. Pairing extremes drives it down, producing a distribution with the tails removed. Neither is the Maxwellian, and Section 6 exhibits both.

5 Necessity without sufficiency

Theorem 4.9 gives a condition whose violation certifies that σ is not invariant. It is natural to hope that its satisfaction certifies the opposite, so that a scheme could be cleared by a single inexpensive test. It does not, and the reason is that the fourth moment is the first member of a family whose later members carry independent information.

Lemma 5.1. *With $Q_6(v) = \sum_i v_i^6$, and angles drawn independently and uniformly,*

$$\mathbb{E}_\theta[Q_6(Tv) \mid v] = \frac{5}{8} \left(Q_6(v) + 3 D(P, v) \right), \quad D(P, v) = \sum_{(a,b) \in P} v_a^2 v_b^2 (v_a^2 + v_b^2).$$

For an exchangeable rule the conditional expectation of D given v is $\overline{D}(v) = (N Q(v) - Q_6(v)) / (N - 1)$.

Proof. A pair contributes $R^6(\cos^6 \theta + \sin^6 \theta)$, and $\mathbb{E}[\cos^6 \theta] = \binom{6}{3} 2^{-6} = 5/16$, so the conditional expectation of the contribution is $\frac{5}{8} R^6$. Expanding $R^6 = v_a^6 + v_b^6 + 3v_a^2 v_b^2 (v_a^2 + v_b^2)$ and summing over pairs gives the identity. For $\overline{D}$, exchangeability gives $\mathbb{E}[D \mid v] = (2(N - 1))^{-1} \sum_{i \neq j} (v_i^4 v_j^2 + v_i^2 v_j^4) = (N - 1)^{-1} \sum_{i \neq j} v_i^4 v_j^2$, and $\sum_i v_i^2 = N$ turns this into $(N Q - Q_6) / (N - 1)$. $\square$

Substituting $\overline{D}$ into Lemma 5.1 and using (3) returns $\mathbb{E}_\sigma[Q_6]$, as Proposition 2.3 requires. For $N \geq 6$ the functional D is not a function of C , so the two conditions are logically independent, and there is no reason to expect one to imply the other. At $N = 4$ they are not independent at all.

Proposition 5.2 (collapse at four particles). *Let $N = 4$. Then for every $v \in \mathcal{S}$ and every matching P ,*

$$D(P, v) - \overline{D}(v) = N(C(P, v) - \overline{C}(v)). \quad (8)$$

Consequently a pairing rule satisfies $\mathbb{E}_\sigma[D] = \mathbb{E}_\sigma[\overline{D}]$ if and only if it satisfies $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$: at four particles the sixth-moment condition carries no information beyond the fourth.

Proof. Write $x_i = v_i^2$. Let P and P' be matchings agreeing except on four indices, with $(a, b), (c, d) \in P$ and $(a, c), (b, d) \in P'$. The common terms cancel, leaving

$$C(P) - C(P') = x_a x_b + x_c x_d - x_a x_c - x_b x_d = (x_a - x_d)(x_b - x_c).$$

Collecting the same four terms of D ,

$$\begin{aligned} D(P) - D(P') &= x_a x_b (x_a + x_b) + x_c x_d (x_c + x_d) - x_a x_c (x_a + x_c) - x_b x_d (x_b + x_d) \\ &= (x_b - x_c) [(x_a^2 - x_d^2) + (x_b + x_c)(x_a - x_d)] \\ &= (x_a - x_d)(x_b - x_c)(x_a + x_b + x_c + x_d), \end{aligned}$$

so that

$$D(P) - D(P') = (C(P) - C(P'))(x_a + x_b + x_c + x_d). \quad (9)$$

At $N = 4$ there are exactly three matchings, any two of them differ by such an exchange, and the four indices it touches are all of them, so the multiplier in (9) is $\sum_i x_i = N$. Hence $D(P) - NC(P)$ takes one value at all three matchings. Averaging that constant over the three returns $\overline{D} - N\overline{C}$, and subtracting gives (8). Since (8) holds for every matching it holds for the matching a rule produces, and taking expectations under σ gives $\mathbb{E}_\sigma[D] - \mathbb{E}_\sigma[\overline{D}] = N(\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\overline{C}])$, from which the equivalence is immediate. $\square$

The multiplier in (9) is the sum over the four indices the exchange touches, not over the whole system, so the collapse is special to $N = 4$: for $N \geq 6$ that sum varies from one exchange to the next and D is no longer an affine function of C across matchings. The hierarchy of this section therefore begins at $N = 6$.

For $N \geq 6$ the fourth-moment condition is not expected to imply the rest, and under a condition recorded in the proposition below, which is verified by exact computation over a range of N rather than proved in general, the failure can be exhibited exactly.

Proposition 5.3 (the converse of Theorem 4.9 fails). *Let $N \geq 6$ be even. For a pairing rule write δ_C for $\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\overline{C}]$ and δ_D for $\mathbb{E}_\sigma[D] - \mathbb{E}_\sigma[\overline{D}]$, and let $\delta_C^{\text{srt}}, \delta_D^{\text{srt}}$ and $\delta_C^{\text{ext}}, \delta_D^{\text{ext}}$ denote their values for the sorted and extremal rules. If*

$$\delta_C^{\text{srt}} \delta_D^{\text{ext}} \neq \delta_C^{\text{ext}} \delta_D^{\text{srt}}, \quad (10)$$

then there is a state-dependent pairing rule with $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$ exactly, so that the hypothesis of Theorem 4.9 is not met, and for which σ is nonetheless not invariant.

Proof. Let the rule apply the sorted rule with probability p and the extremal rule with probability $1 - p$, the choice drawn independently of the state. Both δ_C and δ_D are expectations, so they are affine in p : $\delta_C(p) = p\delta_C^{\text{srt}} + (1 - p)\delta_C^{\text{ext}}$, and likewise for δ_D . By Corollary 4.10, $\delta_C^{\text{srt}} > 0 > \delta_C^{\text{ext}}$, so

$$p^* = \frac{-\delta_C^{\text{ext}}}{\delta_C^{\text{srt}} - \delta_C^{\text{ext}}} \in (0, 1) \quad \text{and} \quad \delta_C(p^*) = 0$$

exactly, not merely to within a sampling resolution. At that p^* ,

$$\delta_D(p^*) = \frac{\delta_C^{\text{srt}} \delta_D^{\text{ext}} - \delta_C^{\text{ext}} \delta_D^{\text{srt}}}{\delta_C^{\text{srt}} - \delta_C^{\text{ext}}},$$

whose numerator is nonzero precisely under (10). Now integrate Lemma 5.1 against σ , which gives $\mathbb{E}_\sigma[Q_6(Tv)] = \frac{5}{8}(\mathbb{E}_\sigma[Q_6] + 3\mathbb{E}_\sigma[D])$, and replacing $\mathbb{E}_\sigma[D]$ by $\mathbb{E}_\sigma[\overline{D}]$ there returns $\mathbb{E}_\sigma[Q_6]$. Hence $\delta_D(p^*) \neq 0$ forces $\mathbb{E}_\sigma[Q_6(Tv)] \neq \mathbb{E}_\sigma[Q_6]$, and σ is not invariant. $\square$

Condition (10) asks that the sorted and extremal rules move the fourth and sixth moments by non-proportional amounts. It fails at $N = 4$, where Proposition 5.2 makes the two sides equal, which is why the proposition is stated for $N \geq 6$.

Beyond $N = 4$ the condition can be checked exactly. Both sides of (10) are homogeneous in the loads, of degree two in C and degree three in D , so on the sphere the radial factor cancels from the ratio and the constraint disappears: each side becomes an expectation over N independent χ_1^2 variables, evaluated as a two-dimensional integral against the bivariate order-statistic density. The display below reports the ratio $R(N) = \delta_C^{\text{ext}} \delta_D^{\text{srt}} / (\delta_C^{\text{srt}} \delta_D^{\text{ext}})$ so computed. It exceeds unity at every N listed and moves further from it as N grows, so (10) holds throughout; the resulting one-step change in $\mathbb{E}_\sigma[Q_6]$ runs from about two percent at $N = 6$ to fifteen percent at $N = 64$. Proposition 5.5 extends both statements to every even $N \geq 6$, granted the condition displayed there.

N	6	8	10	12	14	16	20	24	32	48	64
$R(N)$	1.124	1.208	1.269	1.316	1.353	1.384	1.433	1.469	1.521	1.584	1.621

Three identities available in closed form serve as controls on the quadrature: $\sum_i \mathbb{E}[y_i] = N$, $\sum_{i < j} \mathbb{E}[y_i y_j] = N(N-1)/2$ and $\sum_{i < j} (\mathbb{E}[y_i^2 y_j] + \mathbb{E}[y_i y_j^2]) = 3N(N-1)$. Their largest relative error is 4.5×10^{-10} for $N \leq 24$, rising to 6.0×10^{-5} at $N = 64$, where the inner quadrature becomes roundoff-limited. Each entry above is quoted to fewer digits than its control error supports.

The same reduction shows why $R(N) > 1$ is not delicate. Writing S_{srt} and S_{ext} for the ratio of third- to second-order contributions under each rule, taken throughout in the independent-load normalisation the reduction supplies, $R = S_{\text{srt}}/S_{\text{ext}}$, and the two rules separate: S_{srt} increases with N , from 8.494 at $N = 6$ to 10.914 at $N = 64$, while S_{ext} decreases, from 7.556 to 6.733. The quantities move apart from a gap of 0.939 already present at $N = 6$. A general proof of (10) would therefore not require comparing the rules at each N ; it would be enough to establish the two one-sided bounds $S_{\text{srt}}(N) \geq S_{\text{srt}}(6)$ and $S_{\text{ext}}(N) \leq S_{\text{ext}}(6)$, each a statement about one rule alone. Proposition 5.5 below establishes both, granted one condition on a plane curve built from the two rules. The normalisation matters for the one-sided statements and not for the ratio: passing from the sphere to independent loads scales S by $N/(N+4)$, so R and (10) are unchanged, while S_{ext} decreases with N in the independent-load normalisation used here and rises in the spherical one.

Proposition 5.3 defeats the condition of Theorem 4.9, which is an equality of σ -expectations. It does not defeat the test as a practitioner applies it, on a single state: the rule takes the sorted branch or the extremal one, so at $N = 64$ it reads $C/\overline{C}$ near 2.9 or near 0.14 and never near unity. Defeating the one-state reading is a separate requirement, and the construction below meets it.

Remark 5.4 (the one-state reading can be passed by a bad rule). Proposition 5.3 defeats an equality of σ -expectations. The test as a practitioner applies it reads $C/\overline{C}$ from a single state, and that reading too can sit near unity for a rule that does not preserve σ . The evidence for this is numerical, and is set out below rather than proved.

Select a fraction f of the particles uniformly at random, pair those among themselves by the sorted rule, and pair the remainder among themselves by the extremal rule. At $f = 1$ the rule is sorted and $\mathbb{E}_\sigma[C] > \mathbb{E}_\sigma[\overline{C}]$; at $f = 0$ it is extremal and $\mathbb{E}_\sigma[C] < \mathbb{E}_\sigma[\overline{C}]$, both by Corollary 4.10. As f increases through its admissible values the difference $\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\overline{C}]$ changes sign in steps of order N^{-1} , so some f^* makes it vanish to that order. Unlike the mixture of Proposition 5.3, this rule applies a single matching at every state, so its one-state reading of $C/\overline{C}$ sits near unity instead of alternating between two values far from it. The two sub-populations distort in opposite directions and their contributions to C cancel, while nothing forces their contributions to any other functional to cancel with them. That last step is an expectation rather than an argument, and what follows is measurement.

Locating f^* precisely is not possible and not necessary. The cross term is estimated from a finite sample and varies by a few percent between draws, so the crossing is determined only to within that resolution, and quantities read off at an interpolated f^* inherit the imprecision, the more so the higher the moment. Twelve independent determinations at $N = 2048$ placed f^* between 0.299 and 0.316, with mean 0.305 and standard deviation 0.006; an earlier report of five determinations gave the narrower interval 0.297 to 0.308, which twelve show to be optimistic, and the wider figure is the one to use. Across those determinations $C/\overline{C}$ fell between 0.94 and 1.03 and $D/\overline{D}$ between 1.4 and 1.6; the stationary sixth moment, a sixth moment of a heavy-tailed law read off at an interpolated crossing, ranged from 2.0 to 3.7 across the original five. The claim of Remark 5.4 is an existence

claim, and the reproducible form of it is this. At $N = 2048$ there are blend fractions near 0.30 at which $C/\overline{C}$ agrees with unity to within the resolution of the diagnostic itself, while $D/\overline{D}$ is about 1.5, the stationary sixth moment is several times the Maxwellian value, and only about half the mass beyond two thermal speeds survives. A test on C alone would clear such a rule.

We close the section with the reduction promised above, which removes N from the integrand.

Write F for the χ_1^2 distribution function and, for two loads placed at $u < v$, set $a = u$, $b = 1 - v$, $s = a + b$ and $d = b - a$. Summing a symmetric function over a rule's pairs is one integral against the probability that two loads at $u < v$ are matched to each other, the remaining $N - 2$ falling where they will. For the sorted rule that probability is $\frac{1}{2}(s^{N-2} + d^{N-2})$, since the pair is adjacent exactly when nothing lies between the two loads and an even number lies below. For the extremal rule it is the chance that as many lie below u as above v , which the Fourier form of the central binomial coefficient writes as the average of ρ^{N-2} over t uniform on $(-\pi, \pi)$, where $\rho = (1 - s) + \sqrt{s^2 - d^2} \cos t$. Each kernel is therefore a mixture of powers r^{N-2} against a measure that does not depend on N . Folding the sign of r , which is legitimate because N is even, leaves two functions α and β on $(0, 1)$ for each rule with

$$S(N) = \frac{\int_0^1 r^{N-2} \beta(r) dr}{\int_0^1 r^{N-2} \alpha(r) dr}.$$

The whole N -dependence now sits in a single power, and what the two rules need of α and β is a single geometric condition on the plane curve they trace:

$$W = \alpha\beta' - \alpha'\beta \text{ does not vanish on } (0, 1), \text{ and } (\alpha, \beta) \text{ turns through less than } 2\pi. \quad (11)$$

Proposition 5.5. *Suppose (11) holds for each rule; it is verified numerically below with margin, and is not proved here. Then S_{srt} increases and S_{ext} decreases with even N , so $R(N) \geq R(6) > 1$ and (10) holds at every even $N \geq 6$. Moreover $R(N)$ increases to $12(J_2 - 1)/(2J_3 - 6) = 1.8045$, where $J_2 = \int_0^1 F^{-1}(p)F^{-1}(1 - p) dp$ and $J_3 = \int_0^1 F^{-1}(p)^2 F^{-1}(1 - p) dp$.*

Proof. At $N = 2$ the matching is unique and σ is preserved by Proposition 2.3, so δ_C and δ_D vanish there and both functions have zero zeroth moment, $\int_0^1 \alpha = \int_0^1 \beta = 0$. Take the even anchors in increasing order and fix one of them, N_0 ; put $\lambda = S(N_0)$ and $\Theta = \beta - \lambda\alpha$. Then $\int_0^1 \Theta = 0$ and $\int_0^1 r^{N_0-2} \Theta = 0$. If Θ kept one sign it would vanish identically, forcing $\beta = \lambda\alpha$ and $W = 0$, which (11) forbids. If it changed sign once, at r_0 say, then $r^{N_0-2} - r_0^{N_0-2}$ would carry the sign of Θ throughout and $\int_0^1 r^{N_0-2} \Theta = \int_0^1 (r^{N_0-2} - r_0^{N_0-2}) \Theta$ could not vanish. So Θ changes sign at least twice. Its zeros are the points where the plane curve (α, β) meets the line of slope λ through the origin. Since W does not vanish, neither does (α, β) , so the polar angle of the curve is strictly monotone; and since the curve turns through less than a full revolution, each of the two rays is met at most once. Hence Θ changes sign exactly twice, at $r_1 < r_2$.

The order is $(+, -, +)$ rather than $(-, +, -)$ because $\Theta(0^+) = \beta(0) - \lambda\alpha(0) > 0$. For the sorted rule $\alpha(0) = 2(J_2 - 1)$ and $\beta(0) = 2(2J_3 - 6)$; since F^{-1} is increasing, $F^{-1}(p)$ and $F^{-1}(1 - p)$ are oppositely ordered, so Chebyshev's sum inequality gives $J_2 < 1$ and $J_3 < \mathbb{E}[y^2] = 3$, and both are negative, with $\beta(0)/\alpha(0) = (2J_3 - 6)/(J_2 - 1) = 6.650$, while $\lambda \geq S(6)$. For the extremal rule $\alpha(0) > 0$ and $\beta(0)/\alpha(0) = 10.849$ while $\lambda \leq S(6)$. Each bound on λ holds at the first anchor by direct evaluation, and at every later anchor by the conclusion already reached at the anchor before it, which is why the anchors are taken in order.

Let $m > N_0 - 2$ and pick c_0, c_1 so that $g(r) = r^m - c_0 - c_1 r^{N_0-2}$ vanishes at r_1 and r_2 . Then $c_1 = (r_2^m - r_1^m)/(r_2^{N_0-2} - r_1^{N_0-2}) > 0$, and $c_0 = r_1^{N_0-2}(r_1^{m-N_0+2} - c_1) < 0$ because $r_2^{m-N_0+2} > r_1^{m-N_0+2}$. The exponents $0 < N_0 - 2 < m$ carry coefficient signs $(+, -, +)$, so g has at most two positive roots

and therefore exactly r_1 and r_2 ; as $g(0^+) > 0$ and g grows, g has the sign pattern of Θ . So $g\Theta \geq 0$ pointwise, and the two vanishing moments let c_0 and c_1 be subtracted for nothing:

$$\int_0^1 r^m \Theta(r) dr = \int_0^1 g(r) \Theta(r) dr > 0.$$

The denominator of $S(N)$ is δ_C up to a positive factor, so by Corollary 4.10 it is positive for the sorted rule and negative for the extremal one at every even $N \geq 4$, and $S(N) - S(N_0)$ carries the sign of that denominator for $N > N_0$, which gives the two monotonicities and $R(N) \geq R(6)$. For the limit, $\alpha(1) = 2(\mathbb{E}[y^2] - 1) = 4$ and $\beta(1) = 2(2\mathbb{E}[y^3] - 6) = 48$ under the sorted rule, so $S_{\text{srt}} \rightarrow 12$; under the extremal rule the mass concentrates on $a = b$, giving $S_{\text{ext}} \rightarrow (2J_3 - 6)/(J_2 - 1)$. $\square$

We have not proved (11). Both parts concern one pair of functions of one variable on a bounded interval, and both hold with margin. For the first, $W > 0$ throughout for the sorted rule and $W < 0$ throughout for the extremal one. On the sorted arm W vanishes linearly at the left end because $\alpha'(0) = \beta'(0) = 0$ exactly, so the margin to quote is one on W/r , which stays above 27 on $(0, 1)$ and tends to 41.69 as $r \rightarrow 0$. For the second, α and β each change sign exactly once, so each curve crosses to the opposite quadrant and picks up a half turn; with endpoint ratios β/α of 6.650 and 12 for the sorted rule and 10.849 and 6.650 for the extremal one, that puts each curve at about 184 degrees, comfortably inside a revolution. The two parts are not independent. Since $\alpha = \beta = 0$ forces $W = 0$, the first already keeps the curve off the origin, so $\theta = \arg(\alpha, \beta)$ has $\theta' = W/(\alpha^2 + \beta^2)$ of one sign, and the turning is monotone. The total is then $\theta(1^-) - \theta(0^+)$, which the endpoint ratios fix only modulo π ; the sign changes counted above pin the branch. Granted the first part, the second reduces to two endpoint limits and two zero counts. The derivatives are available in closed form and agree with central differences to nine figures, and the two endpoint ratios quoted in the proof are one-dimensional integrals carrying the same controls. What remains is an enclosure, not a search. The reduction also lifts the range of the display above, whose two dimensional quadrature is roundoff-limited near $N = 64$: as a one dimensional moment, R is available at any N .

The pair $\{C/\overline{C}, D/\overline{D}\}$ separates every rule we have examined, including this one. Whether some finite collection of such conditions is sufficient, or whether each even moment contributes an independent constraint indefinitely, we do not know; the question is stated as Conjecture 9.3 below.

6 Numerical study

Four experiments follow, each answering one question. All were run through a generator that tags every reported quantity with its sampling class, computes a standard error, repeats the whole experiment under two independent seeds and reports only the digits agreeing between them. Where a quantity supports one significant figure, one is given.

6.1 Does the one-step identity hold in the code?

Everything downstream rests on Lemma 4.2, which is an identity about one collision step. It is worth testing directly rather than inferring it from agreement further along. We draw $v \sim \sigma$, apply a single step under each pairing rule, and compare the measured $\mathbb{E}_\sigma[Q(Tv)]$ against the exact prediction $\frac{3}{4}(Q + 2C)$ evaluated on the same states.

Table 1 reports the result. The identity holds across rules whose cross-term ratios differ by a factor of twenty-four, with a largest relative discrepancy of 0.028% and no $|d/\text{SE}|$ above one. The

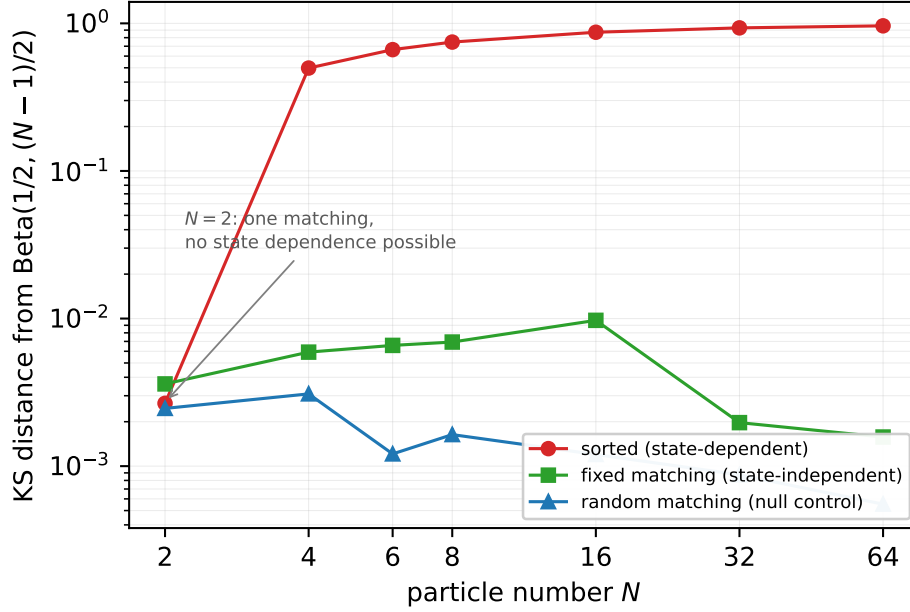


Figure 2: Kolmogorov–Smirnov distance between the empirical law of v_i^2/N and its exact uniform-on-sphere marginal $\text{Beta}(\frac{1}{2}, \frac{N-1}{2})$, after 3000 burn-in steps and 60 collections at intervals of 25, over 400 chains. Random matching is the null control. A fixed state-independent matching sits modestly above it, by a factor between one and a half and eight, which the dependence among its pooled samples accounts for and Proposition 2.3 forbids attributing to bias; the departure of the sorted rule is two to three orders of magnitude larger, so neither determinism nor structure in the pairing breaks the equilibrium. The sorted rule matches the control at $N = 2$, where only one matching exists and no rule can depend on the state, and departs from it at every even $N \geq 4$, as Corollary 4.10 requires. Generated by `kac_manuscript_data_v1.0.0.py`.

two state-independent rules, random and fixed, return $r(v)$ at unity as Lemma 4.7 requires, which is a check on the pairing code independent of the quantity being tested.

The curve rule is not in the table because it is defined on (position, velocity) pairs and a draw from σ carries no positions. Tested instead on the loads of Section 7, where $r(v) = 2.713$ at $N = 2048$ and 2.834 at $N = 8192$, it agrees with Lemma 4.2 to 0.008% and 0.005% respectively. Lemma 4.2 is a statement conditional on v , so it applies to those states as it does to a draw from σ ; what it describes is the collision step alone, and Section 7 composes that step with the projection recorded there.

6.2 Is the failure finite in N ?

Figure 2 answers this. The measure σ is preserved at $N = 2$, where only one matching exists and no rule can depend on the state, and is not preserved at any even $N \geq 4$, with the departure growing to a Kolmogorov–Smirnov distance of 0.96 by $N = 64$. A fixed, deterministic, state-independent matching behaves quite differently. Its Kolmogorov–Smirnov distance rises from 0.0036 at $N = 2$ to 0.0097 at $N = 16$ and falls back to 0.0016 by $N = 64$, sitting between one and a half and eight times the random-matching arm rather than on top of it. That residual elevation is not a transient. The chains are started with an exact draw from σ and Proposition 2.3 keeps them there, so the marginal law is correct at every step and there is nothing to converge to. What the fixed

Table 1: One-step test of Lemma 4.2 at $N = 2048$, 4000 states per rule, angles uniform. The second column is the empirical mean of $C/\overline{C}$ over those states, not a single state’s $r(v)$ and not r_σ , which is a ratio of expectations rather than a mean of ratios; the rules span it from 0.12 to 2.99. Every discrepancy is under 0.03% and within one standard error of zero; a second independent seed agrees. The $w = 512$ row is given to three decimals because two would fall on a rounding tie. Generated by `kac.onestep_test_v1_0_0.py`.

rule	$\widehat{\mathbb{E}}[C/\overline{C}]$	predicted	measured	rel. diff.	d/SE
random	1.002	6137.15	6138.89	+0.028%	+1.00
fixed	0.999	6142.69	6142.70	+0.000%	+0.01
sorted	2.993	9204.53	9202.25	−0.025%	−0.64
extremal	0.122	4792.46	4793.40	+0.020%	+0.73
block, $w = 64$	2.889	9036.08	9038.37	+0.025%	+0.71
block, $w = 512$	2.279	8098.275	8099.592	+0.016%	+0.54

rule loses is independence. It never exchanges partners, so each pair’s energy $v_a^2 + v_b^2$ is fixed for the whole run and the samples pooled across times revisit the same $N/2$ pair energies drawn at the start. The effective number of independent draws is set by the chains and the pairs, not by the collections, and a Kolmogorov–Smirnov statistic computed as though the pooled sample were independent is inflated accordingly. The separation that matters is of a different order altogether. At $N = 64$ the sorted rule stands at 0.96 against the fixed rule’s 0.0016, a factor of six hundred. Neither determinism nor structure in the pairing carries the damage; only dependence on the state does. This is Corollary 4.10 seen in the stationary law rather than in one step.

6.3 Which observables carry the damage?

Table 2 locates it. A point of method belongs here, because it decides what the table is evidence of. After each collision sweep the code applies a projection: it removes the global mean velocity and then rescales to restore the kinetic energy,

$$v \longmapsto (v - \bar{v} \mathbf{1}) \sqrt{\mathcal{E}_0/\mathcal{E}}, \quad \mathcal{E} = \tfrac{1}{2} \|v - \bar{v} \mathbf{1}\|^2,$$

with $\mathcal{E}_0$ the initial kinetic energy. The chain actually simulated is therefore the composition of this map with the collision step, not the collision step alone.

Which results this covers should be stated, because it is not all of them. Table 2, the moment ladder of Section 6, the stationary quantities quoted for the construction of Section 5, and the applied instance of Section 7 all run the projected chain. The finite- N sweep of Figure 2 does not: there the rotation conserves the energy of each pair exactly, the state stays on $\mathcal{S}$ without help, and no projection is applied at any step. The one-step test of Table 1 likewise uses the collision step alone, which is what Lemma 4.2 describes. The second moment is exact to fifteen digits at every block width, including the most damaged, but for two reasons that should be kept apart. The collision conserves the energy of each pair, so the total energy is conserved by the dynamics whatever the pairing rule does, and the projection then enforces it against rounding. The momentum is different: the collision does not conserve it, and it is pinned by the projection alone. In neither case is the agreement evidence about the pairing rule.

Two things follow. The first is that the projection cannot be what produces the damage recorded in Table 2. Sampling the equilibrium of the projected chain directly and comparing with σ gives a

Table 2: Moment ladder across the block width at $N = 2048$, $t = 40$, eight replicates under each of two seeds, normalised so the Maxwellian value is unity. The fourth and sixth moments are heavy-tailed at small w and support one significant figure; the second moment is exact and is given in full. The tail column is not monotone: it rises from 3×10^{-2} at $w = 2$ through 3×10^{-1} to 2×10^0 at $w = 512$, overshooting the Maxwellian value, before settling at 1 for the random rule. The moments below it fall monotonically towards their Maxwellian values across the same row, so the overshoot is a feature of the tail alone. We report it without explanation; a partially aligned rule appears to leave more mass beyond two thermal speeds than either extreme, and we have not established why. The $C/\bar{C}$ column averages over twelve independently drawn equilibrium states, so it estimates the ensemble value; the same quantity in Table 4 is read from one fixed load. Across 600 states the sorted rule gives 2.99 with a load-to-load standard deviation of 0.11, which is the spread a practitioner comparing a single reading against unity must allow for.

w	$C/\bar{C}$	$m_2 = \langle v^2 \rangle$	m_4	m_6	tail
2	3.0	1.0000000000000000	5×10^2	2×10^5	3×10^{-2}
32	3.0	1.0000000000000000	6×10^1	3×10^3	3×10^{-1}
512	2.3	1.0000000000000000	4×10^0	2×10^1	2×10^0
N	1	1.0000000000000000	1×10^0	1×10^0	1.0×10^0

shift in $\mathbb{E}_\sigma[Q]$ of -0.04% at $N = 64$ and -0.0007% at $N = 2048$, diminishing with N and smaller than the fourth-moment error by four orders of magnitude. The second is that the agreement of the conserved moments is not independent evidence of anything, since the projection enforces exactly those moments. Meanwhile the fourth moment is wrong by a factor of order 10^2 and the sixth by 10^5 , and the mass beyond two thermal speeds falls to a few percent of its Maxwellian value.

The consequence for practice is the important part of this section. **A calculation can be wrong in this way while every conservation check it performs reports perfection.** The quantities a particle code monitors as a matter of routine are exactly the ones the projection protects.

6.4 Can it be repaired?

If the low-discrepancy angles supplied the variance reduction and the sorting supplied the bias, one could keep the first and discard the second. Table 3 shows that this fails. Two features of it are worth naming before reading on. The last two rows use the same sorted pairing and differ only in how the angles are supplied, yet their variances differ by seven orders of magnitude. Since the pairing is common to both rows, that gap is attributable to the angles alone. We do not offer a mechanism for its size, and the reader should be slow to take it as a success: it is a spread across replicates, and Section 7 shows in this same setting that a small spread across replicates is compatible with every replicate being wrong by the same large amount. Their biases differ too, by about a third, and that difference is real rather than sampling: it is some fifteen standard errors of the noisier arm. By Lemma 4.2 the two arms have the same one-step drift, since that identity needs only the marginal law of each angle and both supply uniform angles. What differs is the correlation between angles across steps, and the stationary law each arm reaches is evidently not the same. We have not established why the aligned arm settles further from the truth, and we do not claim that alignment worsens the bias in general; the observation is that it does not repair it. Neither difference rescues the scheme. The variance the alignment saves is small beside the bias the

Table 3: The candidate fix. $N = 2048$, 800 steps, 100 randomizations, initial ensemble held fixed. Bias is against the exact equilibrium $\mathbb{E}_\sigma[\langle v^4 \rangle] = 3N/(N+2) = 2.997073$. Biases and MSEs to three significant figures, variances to two.

scheme	bias	variance	MSE
MC (random, iid)	3.03e−3	1.2e−2	1.18e−2
RQMC (random, stratified)	−3.86e−3	1.0e−2	1.02e−2
A-iid (sorted, iid)	1.50e+3	1.2e+5	2.36e+6
Array-RQMC (sorted, aligned)	2.01e+3	1.2e−2	4.06e+6

Table 4: Trade-off across the block width w , aligned angles throughout, 60 randomizations. $w = 2$ is the sorted rule and $w = N$ the random rule. $C/\overline{C}$ here is read from the single fixed load, not averaged over states, so it carries the full load-to-load spread quoted in Table 2. Entries are given to the precision that is reproducible, which is coarser than the precision to which they are determined: the across-randomization standard error of the bias is 1.5×10^{-2} , 1.3, 26, 9.3×10^{-2} and 1.4×10^{-2} down the column. The monotonicity in w is therefore secure even where the printed figures look close, the tightest adjacent margin being about fourteen standard errors between $w = 2$ and $w = 8$.

w	$C/\overline{C}$	bias $\langle v^4 \rangle$	MSE $\langle v^4 \rangle$	bias $\langle v \rangle$	var $\langle v \rangle$	MSE $\langle v \rangle$
2	2.94	2.01e+3	4.06e+6	−7.21e−1	3.6e−9	5.20e−1
8	2.93	2.00e+3	3.99e+6	−7.17e−1	1.3e−6	5.15e−1
64	2.84	7.03e+2	5.34e+5	−6.10e−1	1.4e−4	3.72e−1
512	2.26	9.05e+0	8.24e+1	−3.91e−1	1.1e−5	1.53e−1
2048	0.991	1.57e−2	1.12e−2	−1.80e−4	2.0e−5	1.99e−5

pairing creates, and the mean squared errors of the two rows agree to within a factor of two.

Low-discrepancy angles with random pairing leave the variance of $\langle v^4 \rangle$ where Monte Carlo had it, the two agreeing within their own sampling error. Per-step stratification is washed out by the mixing of the dynamics over 800 steps. What survives the mixing is the sort, which re-imposes low-discrepancy structure on the state at every step, and the sort is what biases the equilibrium.

Table 4 and Figure 3(a) show why no intermediate setting rescues the method either. Across the five widths tested the mean squared error is monotone in the block width for both observables, by margins of fourteen standard errors or more: any amount of velocity-correlated alignment we have tested is net harmful, because the bias it buys always outruns the variance it saves. There is no interior optimum to tune towards.

The variance reduction itself is real but observable-dependent, and is reported here per observable rather than in general. On $\langle |v| \rangle$ full alignment reduces the variance against random pairing by something of order 10^3 , the order Array-RQMC promises; the factor is not reproducible beyond that, being a ratio of two small variance estimates. On $\langle v^4 \rangle$ there is no reduction against Monte Carlo at all. The obstruction rests on the monotonicity, which holds on both observables, and not on the size of any reduction.

6.5 Does the diagnostic work?

Figure 3(b) shows that $r(v)$ does its job: it tracks the measured bias monotonically and, once the rule has produced its matching, costs $O(N)$ operations on one state, before any trajectory is

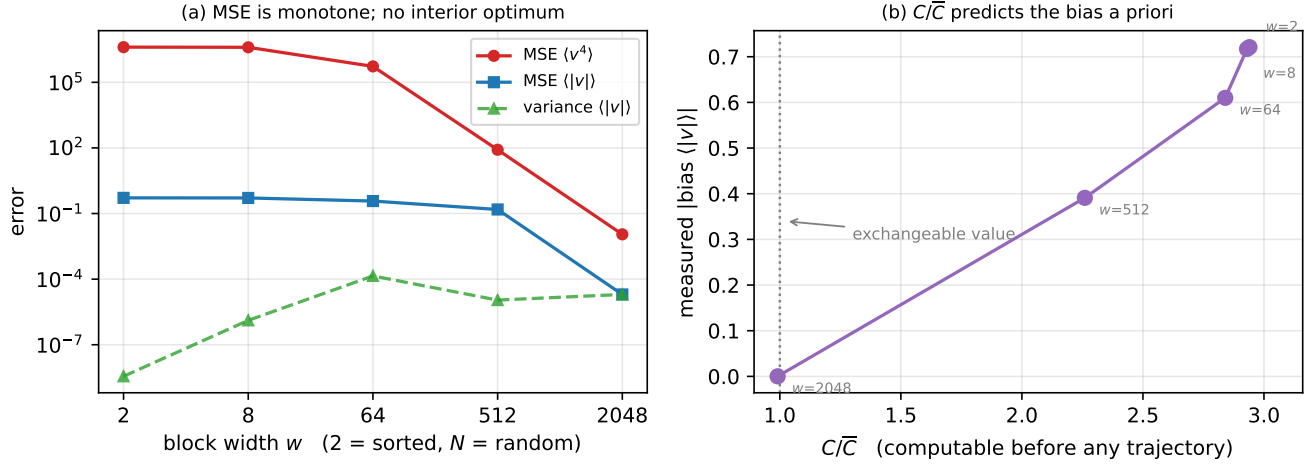


Figure 3: (a) Mean squared error against block width, aligned angles throughout, sixty randomizations. Monotone for both observables: no interior optimum. (b) The cross-term ratio $C/\bar{C}$, computable in $O(N)$ from a single state before any trajectory is run, against the bias subsequently measured. Data generated by `kac_phasef_v2.1.0.py`; plotted by `kac_fig3_tradeoff_v1.0.0.py`.

run. But it is an estimate of r_σ and carries a load-to-load spread, so a tolerance is needed and it must be measured where the comparison is made. Table 2's spread of about 0.11 is that of the sorted rule near $r \approx 3$, which is the wrong regime for a test against unity. Measured instead on a state-independent rule, for which $r_\sigma = 1$ exactly, the spread of $r(v)$ over independent loads is 0.119 at $N = 512$, 0.063 at $N = 2048$ and 0.030 at $N = 8192$, scaling as $N^{-1/2}$ with $\sqrt{N} \text{sd} \approx 2.8$. A practical rule follows: treat a single reading as consistent with unity when it lies within $5.6 N^{-1/2}$ of it, two standard deviations, and read a departure as real only beyond that. The inference also runs one way only. A value of $r(v)$ far from unity is evidence that $r_\sigma \neq 1$ and so condemns a scheme; a value near unity clears it only of the fourth-moment obstruction, by Proposition 5.3, and does not by itself establish $r_\sigma = 1$.

6.6 What the stationary state looks like

The condensate has been named several times above and never shown, so Figure 4 shows it, measured at $N = 2048$ and pooled over fifty chains and two seeds.

Panel (a) gives the stationary density of a single velocity component under the three rules. Random pairing is state independent, so it must return the equilibrium, and it does: the fourth moment reads 3.00 against an exact 2.997, the sixth 14.9 against 14.96, and the mass beyond two standard deviations 0.0455 against 0.0455. That agreement is what licenses reading the sorted arm as signal rather than as an artefact of the sampler. The sorted arm is a spike. Almost all of its mass sits within one bin of the origin, the shoulder is emptied, and the density at the origin runs more than an order of magnitude above the equilibrium.

Panel (b) says where the energy went, and it says so in two numbers. The survival function of the sorted rule drops at once to a plateau near $2/N$ and holds it out to $|v| = \sqrt{N}$, where it stops. Those two readings are the whole description: about two particles in N are moving at all, and between them they carry very nearly the entire energy budget. The rest are at rest.

This also settles what looks at first like a contradiction between two of the measured numbers. The sorted rule retains only 2.1% of the equilibrium mass beyond two standard deviations, which

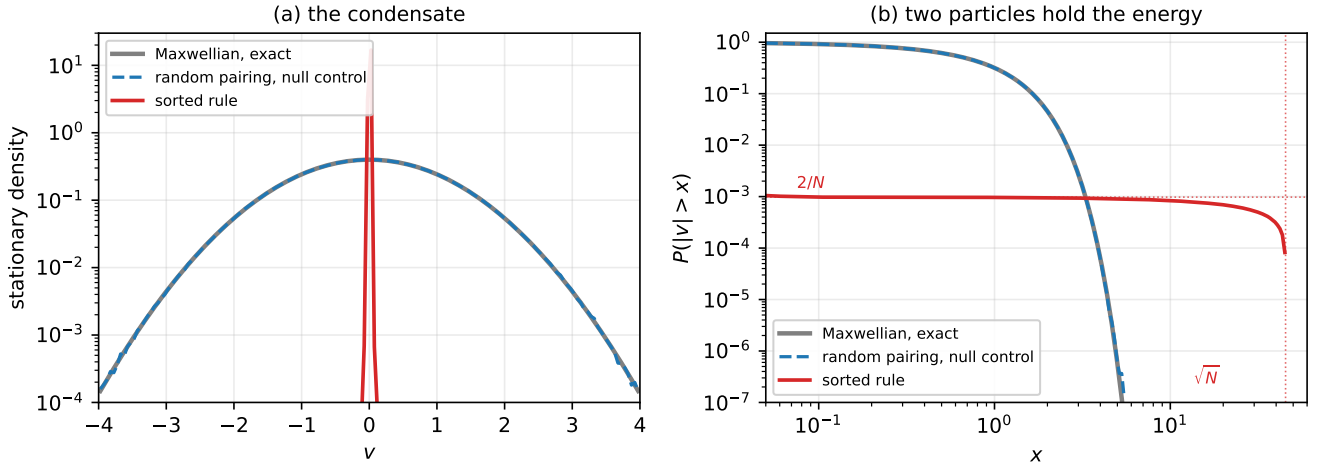


Figure 4: The stationary state of the sorted rule at $N = 2048$, pooled over fifty chains and two seeds. (a) Density of a single velocity component, semi-logarithmic. The sorted rule concentrates almost all of its mass at the origin and empties the shoulder, while random pairing, which is state independent, lies on the equilibrium. (b) Survival function $P(|v| > x)$, logarithmic on both axes. The sorted rule falls at once to a plateau near $2/N$ and holds it to $|v| = \sqrt{N}$: about two particles in N are moving, and they carry the energy. Generated and plotted by `kac_fig4_condensate_v1.0.0.py`.

reads as a thinner tail, while its fourth moment is about 1540 against 3 and its sixth about 2.6×10^6 against 15. Both are the same fact seen twice. Panel (b) shows that survival curve running below the equilibrium through the near tail, where the two-sigma mass is lost, and crossing far above it beyond $|v| \approx 3$, where a plateau of height $2/N$ at distances of order $\sqrt{N}$ supplies moments that the equilibrium cannot. A tail deficit and a moment blow-up are not competing diagnoses of this state, but what a single condensate looks like from two directions.

The measurements above describe the stationary law at one value of N . Section 9 proves that the drain they show is monotone and that it is total.

7 The applied instance

7.1 The test and what it measures

The problem that prompted this work is a one-dimensional collisional particle-in-cell test: charged particles on a periodic domain with a grid-based field solve and a binary collision operator of the kind treated above, run to a warm equilibrium and compared against it through a kernel discrepancy between the empirical joint law of (x, v) and the target. The implementation follows Array-RQMC practice, ordering the particles by the index of a Hilbert curve evaluated at $(x, \Phi(v/v_{\text{th}}))$ and pairing adjacent ranks, which is the curve rule of Section 3.

One thing this test does not do should be said at the outset. Its collision operator is the Kac rotation of Section 2, with the scattering angle drawn uniformly and every selected pair colliding; it is not a physical cross section. The section therefore shows the obstruction reaching a production-shaped code with a grid, a field solve and a real particle load, and it does not show the obstruction escaping the caricature. Those are different claims and only the first is made here.

That rule is not the sorted rule of Corollary 4.10, so the corollary does not apply to it directly,

Table 5: The implementation’s own sorting rule, evaluated on its own Maxwellian load. Two of these columns are predictions and two are measurements, and they should not be read alike. $r(v)$ is deterministic given the load, and the fourth-moment column is the value Lemma 4.2 assigns to that $r(v)$: computed, not observed, and a one-step quantity read from a single state rather than the σ -expectation r_σ that Theorem 4.9 constrains. Table 1 is where that prediction is tested against a measured ensemble. The observed discrepancy is a window mean over $[20, 40]$ as a ratio to the Monte Carlo floor, and is a long-time quantity on the projected chain of Section 6. One load supports no inference about r_σ ; what the row shows is that a load whose $r(v)$ is far from unity behaves as Lemma 4.2 says a state with that $r(v)$ should.

N	$r(v)$ measured	predicted $\Delta Q/Q$, one step from Lemma 4.2	observed discrepancy ratio measured	absolute measured
2048	2.713	+43%	35.0	0.632
8192	2.834	+46%	72.3	0.652

and Theorem 4.9 constrains an expectation under σ that no single load determines. What one load does determine is the drift of the first step, because Lemma 4.2 is conditional on the state. Table 5 evaluates it on the implementation’s own Maxwellian load: the cross term exceeds its exchangeable value by a factor near three, so the fourth moment must rise by some forty percent on the first collision step. Whether the equilibrium itself is displaced is addressed below by measurement rather than by the theorem.

The observed behaviour matches. The discrepancy sits at 35 and 72 times the Monte Carlo floor, and because the ratio grows as $\sqrt{N}$ while the absolute discrepancy does not, the error is independent of the particle number: adding particles does not help. The velocity distribution reached is the condensate that a raised cross term produces, with mass accumulating near zero and a thin population carrying the energy.

7.2 Why the failure is silent

Two features explain why this survived a good deal of investigation. The first is Table 2: the conserved moments are pinned, so the routine checks pass. The second is subtler. Across independent randomizations the discrepancy ratio at $N = 2048$ came back as 35.21, 34.82 and 34.99, a spread of about half a percent, while the correct arm’s ratios spread by nearly thirty percent. The broken arm looks the more converged of the two. It is not. Its error is of order a hundred times the spread of its own replicates; the exact multiple should not be read closely, since a standard deviation from three replicates is itself a coarse quantity, and the point does not need it. Every replicate agrees closely on the same wrong answer, and no amount of replication will reveal the problem. **The tight agreement is the symptom.**

8 Which sorting strategies remain admissible

Theorem 4.9 indicts rules that bias C and is silent about the rest, and the silence is informative. The obstruction is not that a scheme sorts, but that it sorts on the velocities.

Corollary 8.1. *Let the rule order or group the particles by a key η , a random variable on the same space as the state, and suppose η is independent of v under σ . Then the matching P it produces is independent of v , the rule is state-independent in the sense of Definition 2.1, and σ is preserved.*

Independence, not the absence of a functional dependence, is what the argument needs, and the distinction is the one measured below. A key computed from the particle positions is a function of the state and is still admissible when position and velocity are independent under the law at hand; the same key fails when they are not. Whether η is independent of v is therefore a property of the key together with the state it is applied to, and it is checkable rather than assumable.

Proof. Immediate from Proposition 2.3: the matching is then drawn from a distribution not depending on v . $\square$

This covers the cell rules of Section 3, and with them the binary-collision schemes in standard use [18], which group by spatial position and match at random within a group. At equilibrium the velocities carry no information about position, so such schemes are admissible, and the theorem says why. It also explains why the difficulty has not been encountered before: a scheme that groups spatially gains nothing from velocity alignment and has never been asked to provide it.

The comparison with the setting where Array-RQMC has done best is instructive. In the walk on spheres solvers of Ho and Owen [7, 8] the walkers are independent. Each trajectory is a separate sample, and the sort decides only which low-discrepancy point a walker receives, leaving the law of every individual walk untouched. Sorting there can be as aggressive as one likes, and with a Hilbert sort it buys the large reductions those papers report. In a collision model the walkers are not independent. The sort decides which particles collide with which, so it enters the dynamics rather than the assignment of randomness, and the ordering that aligns the ensemble is the ordering that correlates collision partners. The distinction worth carrying away is therefore not between problems where Array-RQMC works and problems where it does not, but between sorts that assign randomness to independent trajectories and sorts that rewire an interaction.

In practice one wants to check a particular implementation rather than argue about it, and $r(v)$ is the test, standing in for the r_σ that Theorem 4.9 constrains. Here a point of definition matters. The exchangeable value (6) averages over *all* matchings, whereas a rule that groups particles can only produce matchings within groups. That restriction is harmless in expectation: Lemma 4.7 gives $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$ for every state-independent rule, and by Corollary 8.1 a grouping key carrying no velocity information is state-independent, so such a rule should read one against the global reference with no dependence on the occupancy.

Measured on the loads of Section 7, a cell rule keyed on position instead reads 0.94 ± 0.01 at occupancy 32 and 0.98 ± 0.01 at occupancy 128, the quoted uncertainty being the spread across twelve independent seeds rather than the scatter within any one of them. The gap is not an artefact of the reference. Every cell in these loads holds an even number of particles, 32 and 128 respectively, so nothing goes unpaired and no occupancy effect arises. Two controls locate the cause. Replacing the position key by a random assignment of the same particles to cells of the same size returns 1.00 ± 0.01 , and keeping the position key while permuting the velocities among the particles returns 1.00 ± 0.01 ; both sit at unity at both particle numbers, as Lemma 4.7 requires. What the diagnostic is detecting is a correlation between position and velocity in the load itself.

The lesson is a practical qualification of Corollary 8.1. The corollary is a statement about the key, and position is velocity-free only in a load where the two are uncorrelated. In a plasma that has been evolved, or loaded with any spatial structure in its temperature, a cell index carries velocity information and the rule keyed on it is state-dependent after all. Whether a key is velocity-free is a property of the key together with the state it is applied to, and it should be measured rather than

assumed. Against the within-group reference

$$\overline{C}_{\text{cell}}(v) = \sum_c \frac{(\sum_{i \in c} v_i^2)^2 - \sum_{i \in c} v_i^4}{2(n_c - 1)}$$

the same configurations return 1.00 ± 0.01 throughout. The two references answer different questions and both are worth computing. The within-group value conditions on the grouping, so it detects bias introduced *inside* the groups and is blind to whatever velocity information the grouping itself carries. The global value sees both together. A rule that reads one against the first and below one against the second, as this one does, is telling the practitioner that its pairing is innocent and its grouping is not.

These measurements are of one cell rule, at two particle numbers and two cell counts. They are consistent with Corollary 8.1 and were run to check it, not to establish a general claim about cell-based schemes: what licenses those is the corollary, and what the corollary requires is that the grouping key carry no velocity information. A scheme whose cells correlate with velocity, as they may away from equilibrium or under a driven load, is not covered.

The two cautions of Section 6 apply here unchanged: the test is one-sided, and $r(v)$ is a sample quantity whose load-to-load spread sets the smallest difference worth reading.

9 The energy cascade and open problems

Theorem 4.9 fixes the direction in which the first collision moves $\mathbb{E}_\sigma[Q]$ and says nothing about where the chain settles. For the sorted rule one elementary thing can be said about that. Order the squared speeds, $v_{(1)}^2 \leq v_{(2)}^2 \leq \dots \leq v_{(N)}^2$, and let

$$\mathcal{E}_j(v) = \sum_{i=1}^j v_{(i)}^2, \quad 0 \leq j \leq N, \quad (12)$$

be the energy held by the j slowest particles, so that $\mathcal{E}_0 = 0$ and $\mathcal{E}_N = N$.

Theorem 9.1 (monotone energy cascade). *Let N be even and let the pairing be the sorted rule, with ties in speed broken in any way. Let each collision rotate its pair through an angle of any law, shared across pairs or not. Then for every even j the energy $\mathcal{E}_j$ of the j slowest particles is non-increasing along every trajectory.*

Proof. Fix a state and one step. Order the particles by speed, breaking ties in any way, so that the j slowest are ranks 1 to j . Since j is even they form the pairs $(1, 2), (3, 4), \dots, (j-1, j)$, which are $j/2$ of the pairs the sorted rule creates. A collision rotates its pair on the circle of radius $(v_a^2 + v_b^2)^{1/2}$ and conserves that pair's energy whatever the angle, so these same j particles hold exactly $\mathcal{E}_j$ after the step. The j smallest squared speeds after the step sum to no more than the squares of any j particles, these included. Hence $\mathcal{E}_j$ cannot rise. The tie-break plays no part: $\mathcal{E}_j$ depends only on the multiset of squares and any ordering leaves the j slowest a union of whole pairs. $\square$

For odd j the argument fails at its first step. Rank j is paired with rank $j+1$ across the boundary and that pair's energy can flow down into rank j . It does, and by a lot. The theorem is a one-step statement, so it was tested one step at a time from states drawn to cover the space: uniform on $\mathcal{S}$, two-particle condensates with the remainder holding 10^{-2} or 10^{-6} of the energy, every speed tied, speeds tied in random index pairs, and magnitudes spread over eight decades,

with twenty thousand states per family and two seeds at each N , and two hundred trajectories of three hundred steps besides. For every even $j < N$ and every N from 4 to 256 the largest one-step increase of $\mathcal{E}_j$ over all of it was at most four units in the last place of N , which is the rounding floor of double precision. For the control $j = 3$ the largest increase was 2.00 at $N = 4$, the boundary pair's whole energy, and 0.12 at $N = 16$. The theorem rests on the pairing structure and not on the arithmetic.

The cascade says which way energy moves. $\mathcal{E}_j$ is non-increasing and bounded below by zero, so along every trajectory it falls to a limit $L_j \geq 0$ that is non-decreasing in j . Energy drains upward through the ranking and never returns; the top pair at ranks $N - 1$ and N holds $N - \mathcal{E}_{N-2}$ and can only gain. That is the mechanism of the condensate Figure 4 shows, and Theorem 4.9 detects the same concentration in the mean on the first collision as a fourth-moment drift. Whether the drain is total is a separate question, and the answer is yes.

Theorem 9.2 (the condensate). *Let $N \geq 4$ be even and fix any initial state. Take the sorted rule, with the collision angles drawn independently across steps; as in Theorem 9.1, the angle within a step may be shared across pairs or not.*

(i) Condensation. *Suppose the scattering law has a density bounded below on an interval longer than a quarter turn, uniformly in the pre-step state. Then $\mathcal{E}_{N-2} \rightarrow 0$ almost surely: the two largest squared speeds carry the whole energy in the limit, on the circle of radius $\sqrt{N}$, and the $N - 2$ smallest tend to zero. The statement is about ranks. It does not assert that any fixed pair of particles eventually carries the energy, and in general none does.*

(ii) The limiting phase. *Under the uniform scattering law, which satisfies the hypothesis of (i), order the two particles of the top pair by index rather than by speed and let Φ be the phase of the resulting ordered pair. Then Φ converges in total variation to the uniform law. The ordering matters: ordered by speed, Φ is confined to the half of the circle on which $|\cos| \leq |\sin|$ and cannot converge to the uniform law.*

Proof. Write E_n for $\mathcal{E}_{N-2}$ after n steps. By Theorem 9.1 it is non-increasing and bounded below by zero, so it falls to a limit $L \geq 0$ with $E_n \geq L$ for every n . It suffices to show $\mathbb{P}(L \geq \ell) = 0$ for each $\ell > 0$.

Fix $0 < \ell \leq N$, there being nothing to prove for larger ℓ , and a step at which $E_n \geq \ell$. The two fastest particles form the top pair and hold $T = N - E_n$; the other $N - 2$ form whole pairs and hold E_n , before the step and after it. After the step the top pair's squares are $T \cos^2 U$ and $T \sin^2 U$, where U is the pair's phase after the step: the collision adds an angle drawn from the scattering law, so conditionally on the state before the step U is distributed as that law translated by the pre-step phase, and under the uniform law U is uniform on $[0, 2\pi)$. Let M be the largest square among the other $N - 2$ particles after the step, so that $M \geq E_n/(N - 2) \geq \ell/(N - 2)$, and let $a = T \min(\cos^2 U, \sin^2 U) \leq N \min(\cos^2 U, \sin^2 U)$. The two largest squares after the step sum to at least $T - a + M$, since $T - a$ and M belong to different particles, so

$$E_n - E_{n+1} \geq M - a \geq \frac{\ell}{N - 2} - N \min(\cos^2 U, \sin^2 U).$$

Put $\eta = \ell/(2N(N - 2))$ and $\delta = \ell/(2(N - 2))$. If $\min(\cos^2 U, \sin^2 U) \leq \eta$ the decrease is at least δ . That event is a union of arcs about the four multiples of a quarter turn, each of half-width $\arcsin \sqrt{\eta}$. An interval longer than a quarter turn contains such a multiple with at least $d/2$ of room on each side, where the hypothesis of (i) supplies a density bound $c > 0$ on an interval of length $\pi/2 + d$. The event therefore has probability at least $p \geq 2c \min\{\arcsin \sqrt{\eta}, d/2\} > 0$, fixed by ℓ , N and the law, and not by the state. Under the uniform law $p = (4/\pi) \arcsin \sqrt{\eta}$.

Let A_n be the event that $E_{n+1} \leq E_n - \delta$, and let $\mathcal{F}_n$ be generated by the first n steps. The bound above gives $\mathbb{P}(A_n \mid \mathcal{F}_n) \geq p$ on $\{E_n \geq \ell\}$, an event in $\mathcal{F}_n$, and $\{L \geq \ell\} \subseteq \{E_n \geq \ell\}$ for every n , so on $\{L \geq \ell\}$ the sum $\sum_n \mathbb{P}(A_n \mid \mathcal{F}_n)$ diverges. By Lévy's extension of the Borel–Cantelli lemma [19, §12.15], A_n then occurs infinitely often almost surely on $\{L \geq \ell\}$. But E_n lies in $[0, N]$ and each occurrence removes at least δ , so A_n can occur at most N/δ times. Hence $\mathbb{P}(L \geq \ell) = 0$, and $L = 0$ almost surely.

The limit describes the state. Write e for the pre-step value of $\mathcal{E}_{N-2}$ and $T = N - e$ for the energy of the pair the pre-step ranking selects; the $N - 2$ slowest carry e , and the top pair carries the rest on the circle of radius $T^{1/2}$.

For (ii), a step rotates the pair the *pre-step* ranking selects, and under the uniform law that pair leaves the step with phase exactly uniform and independent of the state before it. The pair the *post-step* ranking selects may differ, and whether it does depends on the rotation, so the two must not be identified. Let M be the largest square among the particles outside the rotated pair after the step. Those particles form whole pairs, so the step conserves their total energy, which is the pre-step value e ; hence $M \leq e$. The post-step value of $\mathcal{E}_{N-2}$ equals e off the displacement event D and is strictly smaller on it, which is how Theorem 9.1 makes it fall; e and T below are always the pre-step values. The rotated pair leaves the top two only if its smaller square $T \min(\cos^2 U, \sin^2 U)$ falls below M , its larger square being at least $T/2 > M$ once $e < N/3$. Since M is fixed by the other pairs, the estimate is conditional on it as well as on the past:

$$\mathbb{P}(D \mid \mathcal{F}_n, M) = \frac{4}{\pi} \arcsin \sqrt{M/T}, \quad \text{hence} \quad \mathbb{P}(D \mid \mathcal{F}_n) \leq b(e),$$

where $b(e) = (4/\pi) \arcsin \sqrt{e/(N - e)}$ for $e < N/3$ and $b(e) = 1$ otherwise, so that $b \leq 1$ in every state. Off D the post-step top pair *is* the rotated pair, and the index ordering fixed in the statement does not depend on which of the two is faster, so the two phases agree there. Coupling them, the conditional total variation distance between the law of Φ and the uniform law is at most $b(e)$. Since $e \rightarrow 0$ almost surely by (i) and $b \leq 1$, bounded convergence gives the unconditional convergence asserted in (ii). $\square$

Almost surely cannot be improved to every trajectory, since a trajectory of zero rotations stalls. The theorem gives no rate; in simulation the decay is polynomial rather than geometric, and at $N = 16$ from uniform states a strict decrease of $\mathcal{E}_j$ occurs with probability between 0.85 and 0.99 for even j and removes between a fifth and three fifths of $\mathcal{E}_j$ on average.

The stationary law is therefore arithmetic. A single component is $\sqrt{N} \cos \Phi$ with Φ uniform, or zero, and the even moments per particle are

$$m_{2k} = \frac{2}{N} N^k \mathbb{E}[\cos^{2k} \Phi] = \left(\frac{N}{2}\right)^{k-1} \frac{(2k-1)!!}{k!}, \quad m_4 = \frac{3N}{4}, \quad m_6 = \frac{5N^2}{8},$$

against $3N/(N+2)$ and $15N^2/((N+2)(N+4))$ under σ , which tend to 3 and 15. The fourth moment of the sorted rule's stationary state grows linearly in N and the sixth quadratically. Measured for the sorted rule at $N = 4, 16, 64, 256, 1024$ and 2048 under two seeds, m_4 sits within -0.5% to $+1.1\%$ of $3N/4$ and m_6 within -0.9% to $+2.0\%$ of $5N^2/8$, deviations comparable to the two-seed spread. The fastest particle carries $\max_i |v_i|/\sqrt{N} \geq 0.999999$ at every N . Theorem 9.2(ii) derives the law; measurement confirms it over a 512-fold range in N .

Remark (what each part needs). The density in (i) cannot simply be dropped, though we show this by measurement rather than by proof. An atom at an irrational rotation is aperiodic, so its phase enters every neighbourhood of the circle, but it carries no density and the conclusion fails.

At $N = 16, 64$ and 256 under two seeds, the energy below the top pair holds between 8 and 12 percent of the total and does not vanish. An atom at a quarter turn is periodic and fails further, its phase confined to four points.

Part (ii) is not claimed beyond the uniform law, and measurement does not invite it. Across three further laws satisfying (i), two arcs of length a quarter turn plus 0.30 and plus 0.05, and a floored non-uniform density, the measured m_4 and m_6 stay within 1.7 and 3.0 percent of $3N/4$ and $5N^2/8$ without matching them. For two of the three the deviations are one-signed at every N and both seeds, and fall wholly outside the 0.4 and 0.7 percent the uniform law returns under the same protocol; the third is indistinguishable from it. Six measurements to a law is thin evidence for a systematic effect, so the bands are reported and nothing more. The identity in (ii) uses the uniform phase exactly, and appears particular to it.

Conjecture 9.3 (length of the hierarchy). Is there a finite collection of moment conditions whose simultaneous satisfaction implies that σ is preserved? The construction of Proposition 5.3 defeats the fourth-moment condition alone using one free parameter, and a construction with k free parameters might be expected to defeat any k conditions. Proposition 5.2 shows the hierarchy is degenerate at $N = 4$, where the sixth-moment condition is implied by the fourth, so a second question is where it begins.

Conjecture 9.4 (how much velocity information a sort may carry). Grouping the particles into N/w blocks by speed conveys $\log_2(N/w)$ bits per particle about the velocities, while a full ordering conveys $\log_2 N$. At $N = 2048$ we find one bit admissible and two not, against the eleven a full ordering supplies. Whether the admissible budget is a constant number of bits, a constant number of groups, or some fraction of $\log_2 N$ cannot be determined from a single particle number.

Finally, the model treated here is the Kac caricature. Lemma 4.5 removes the restriction to uniformly resampled scattering angles, but the velocities remain one-dimensional and every pair collides at every step. Whether the obstruction survives in three dimensions, with variable collision rates and a physical cross section, is the question that decides how far the conclusion reaches. A second limit is narrower but nearer to practice. Corollary 8.1 clears any key carrying no velocity information, and that is a statement about the key together with the state it is applied to, not about the key alone: in a load where position and velocity are correlated, a spatial key carries velocity information and the guarantee lapses. Section 8 measures exactly that happening. A practitioner relying on the corollary should therefore measure whether the chosen key is velocity-free on the state at hand rather than assume it from the key's definition.

10 Conclusion

We began with a method that works and a problem that ought to suit it. Array-RQMC earns its variance reductions by sorting trajectories and handing them low-discrepancy points in that order, and collisional kinetics offers long chains, low per-step dimension, and an average over many particles. The obvious sorting key is the velocity, because the velocity is what the collision operator acts on. Coulibaly and Lécot took that route in 1997 and recorded that in more than one dimension the choice of ordering could decide whether the method helped, without a reason why.

That is the key the adaptation cannot safely use, once the order it induces is what selects the collision partners. Averaged over the scattering angle, the pairing enters the one-step fourth moment through one scalar (Lemma 4.2), and that scalar sits at a definite value for any rule chosen independently of the state. A rule that moves it away from that value moves the equilibrium with

it, at every even $N \geq 4$ and under any symmetric scattering law shared by all pairs, provided that law is not concentrated on the quarter turns that leave Q fixed. The sorted and extremal rules are proved to move it, and the block and curve rules are measured to. The alignment that buys the variance reduction and the correlation that destroys the equilibrium are not two effects to be balanced against each other. They are one structure seen twice, which is why Figure 3(a) shows no interior optimum: there is nothing to tune.

Three things follow that we would want a practitioner to take away. Sorting is not the difficulty; sorting on velocity is, and a grouping key carrying no velocity information leaves the equilibrium intact. The cross term can be computed in one pass before any trajectory is run, so a proposed scheme can be tested in advance rather than diagnosed afterwards. That test is one-sided and reads a single state: a value far from the exchangeable one condemns a scheme, a value close to it clears only the fourth-moment obstruction, and neither is the σ -expectation the theorem constrains. And the failure does not announce itself: the conserved moments stay exact, so the usual checks pass, while the replicates agree with one another to a fraction of a percent around an answer that is wrong by many standard deviations. In this setting the close agreement of independent runs is not evidence that a computation has converged. It is the signature of the thing that has gone wrong.

Where the chain goes instead is now settled. For the sorted rule the energy of the slowest particles can only fall, so the chain drains energy upward through the ranking at every collision and along every path (Theorem 9.1), and for any scattering law with a density spread over more than a quarter turn the drain is total (Theorem 9.2): almost surely the two fastest end with everything, moving on a circle of radius $\sqrt{N}$, and the rest are at rest, though the two need not be the same particles from step to step. That is the mechanism of the condensate Section 6 draws, and under the uniform law its stationary law follows by arithmetic and matches measurement over a 512-fold range in N .

Data and code availability

The code that produces every figure, table and reported number, together with the generated data and the particle loads used in Section 7, is available at <https://github.com/machyman/hyman2026velocity>. Each generator has a notebook wrapper that runs it in Colab without a local installation. The repository records which artifacts are regenerated by a deposited script and which are authored by hand, and carries the checkers used to verify both.

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